Immuno-Oncology & Checkpoint Inhibitors

Anti-PD-1, Anti-CTLA-4, TMB, MSI, and Immune-Related Adverse Events

Doc #05 of 15	**Domain:** Oncology & Cancer Treatment	**Audience:** Medical Professionals

Keywords: immunotherapy | PD-1 | PD-L1 | CTLA-4 | TMB | MSI-H | irAE | checkpoint inhibitor

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1. The Cancer Immunity Cycle

Chen and Mellman's cancer immunity cycle describes the seven steps required for effective anti-tumour immunity: (1) release of tumour antigens, (2) antigen presentation by DCs, (3) T-cell priming/activation, (4) T-cell trafficking to tumour, (5) infiltration, (6) recognition of cancer cells by T cells, (7) tumour cell killing. Checkpoint pathways operate primarily at steps 3 and 6, providing therapeutic windows for inhibition.

2. Immune Checkpoint Biology

Checkpoint	Ligand(s)	Location	Primary Effect
CTLA-4	CD80 (B7-1), CD86 (B7-2)	Lymph node (priming phase)	Suppresses early T-cell activation; Treg enhancement
PD-1	PD-L1 (CD274), PD-L2	Tumour microenvironment	T-cell exhaustion; effector function loss
LAG-3	MHC-II, FGL1	Tumour microenvironment	Co-inhibition with PD-1 on exhausted T cells
TIM-3	Galectin-9, HMGB1	Tumour microenvironment	Terminal exhaustion marker
TIGIT	CD155, CD112	Tumour microenvironment	NK and T-cell suppression

3. Approved Checkpoint Inhibitors

Agent	Target	Key Approvals
Pembrolizumab (Keytruda)	PD-1	NSCLC, HNSCC, TNBC, MSI-H solid tumours, TMB-H (≥10 mut/Mb), Hodgkin HL, Cervical, Endometrial, ESCC, GC, CSCC, urothelial, RCC, HCC, BTC, CRC, MCC, TNBC
Nivolumab (Opdivo)	PD-1	NSCLC, RCC, Melanoma, HNSCC, CRC, HCC, ESCC, GC, urothelial, MPM
Atezolizumab (Tecentriq)	PD-L1	NSCLC (1L+), TNBC, HCC, SCLC
Durvalumab (Imfinzi)	PD-L1	Stage III NSCLC (PACIFIC), SCLC, BTC
Ipilimumab (Yervoy)	CTLA-4	Melanoma (mono + combo), RCC, CRC MSI-H, MPM, NSCLC (+ nivo)
Relatlimab + Nivolumab	LAG-3 + PD-1	Unresectable / metastatic melanoma (1st line)

4. Predictive Biomarkers

- PD-L1 expression (IHC) — TPS (tumour proportion score) and CPS (combined positive score); context and assay-specific thresholds (e.g., pembrolizumab NSCLC: TPS ≥50% for 1L monotherapy)
- MSI-H / dMMR — FDA pan-tumour approval (pembrolizumab); ~4% of all solid tumours but enriched in endometrial, CRC, gastric
- Tumour Mutational Burden (TMB) — ≥10 mut/Mb by FoundationOne CDx; approved for pembrolizumab TMB-H solid tumours (KEYNOTE-158)
- KRAS status — KRAS G12C CRC associated with inferior ICI response; KRAS wt required for cetuximab
- EBV positivity — EBV+ gastric cancer responds better to anti-PD-1
- CD8+ TIL density (exploratory) — inflamed vs. excluded vs. desert phenotype predicts response

5. Immune-Related Adverse Events (irAEs)

irAEs arise from immune activation in normal tissues. Incidence is higher with CTLA-4 inhibitors (~60% any grade) than PD-1/L1 agents (~30%), and highest with combination therapy (~90%). Any organ can be affected; most common are dermatitis, colitis, hepatitis, pneumonitis, and endocrinopathies.

irAE	Grade 3-4 Incidence	First-Line Management	Time to Onset
Colitis	8-22% (ipi)	IV methylprednisolone 1-2 mg/kg/d → infliximab if steroid-refractory	5-10 weeks
Pneumonitis	2-5%	Hold ICI; pred 1-2 mg/kg/d; bronchoscopy if diagnostic uncertainty	2-24 months
Hepatitis	3-10% (combo)	Hold ICI; pred 0.5-1 mg/kg/d; MMF if refractory	6-12 weeks
Hypophysitis	1-8% (ipi)	Hormone replacement; steroids for mass effect	9 weeks (ipi)

Thyroiditis	1-3%	Levothyroxine for hypothyroidism; beta-blocker for thyrotoxicosis	4-7 weeks
Myocarditis	<1% (fatal risk high)	Hospitalise immediately; high-dose IV steroids	3-27 days

■ URGENT: Grade 3-4 irAEs typically require immunotherapy hold and high-dose corticosteroids. Permanently discontinue for grade 4 (except endocrinopathies). Consult relevant subspecialty urgently.